

FAN SI

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Authorized To Work In The U.S

PROFILE SUMMARY

An analytical and performance-driven professional with hands-on experience developing scalable server-side architectures, optimizing distributed systems, and engineering high-fidelity applications across IT, gaming, and cloud environments. Proven success in strengthening system reliability, security, and performance while designing APIs, automating deployment pipelines, and troubleshooting complex backend challenges to deliver secure, maintainable, and business-focused solutions. Committed to continuous learning, technical innovation, and applying expertise to solve complex challenges and deliver measurable results.

PROFESSIONAL EXPERIENCE

IT Engineer

NGEN Technology-Dallas, TX

Aug 2023 - Present

- Administer enterprise infrastructure by configuring Windows and Linux servers, VMware ESXi, Active Directory, Office 365, and endpoint systems to strengthen uptime and operational continuity.
- Manage network architecture across LAN, WAN, VLANs, TCP/IP, DNS, and DHCP, deploying Cisco routers, switches, and firewalls to improve security and optimize connectivity.
- Deploy and sustain IP-based surveillance systems with structured cabling, PoE switches, and video management software to improve facility-wide physical security and centralized monitoring.
- Execute data backup, disaster recovery, and patch management across hybrid environments, leveraging Azure and AWS cloud platforms to mitigate risks and safeguard business-critical assets.
- Diagnose and remediate hardware, software, and connectivity issues using ITIL service management practices, system monitoring tools, and root-cause analysis to improve performance in local and remote data centers.

Back-End Developer

Beijing DaSi Culture & Art Co., Ltd.

Sep 2022 - Mar 2023

- Built scalable server-side architectures and RESTful APIs using Java, strengthening system reliability and performance for high-traffic customer-facing applications.
- Optimized cloud-based distributed systems through load balancing, resource allocation, and latency reduction, elevating application responsiveness across mobile and desktop clients.
- Refined backend functionality by debugging complex issues, profiling performance constraints, and implementing robust solutions that stabilized production environments.
- Structured agile workflows with project managers, QA, and developers, aligning technical roadmaps with client requirements to accelerate secure, maintainable codebases delivery.
- Designed reusable back-end modules, automated testing schedules, and integration pipelines, streamlining cross-browser and multi-device compatibility for seamless user experiences.

UE4 Game Developer

Hero Games (HERO Entertainment Co., Ltd.), China

Feb 2022 - Aug 2022

- Constructed and fine-tuned high-fidelity 3D environments in Unreal Engine 4 to elevate rendering stability and optimize asset streaming.
- Crafted modular UI components and interactive gameplay features using Blueprints and C++, enhancing user engagement.
- Implemented custom engine utilities and automated workflows, reducing iteration cycles by 15% and accelerating build deployment.
- Assembled visual effects and animation logic via Sequencer, Niagara, and Material Editor to align visual fidelity with game mechanics.
- Documented technical pipelines, tool usage, and system structures to support maintainability, onboarding, and multi-team knowledge transfer.

ACADEMIC PROJECTS

Third-Person Battleship Shooting Game

Oct 2021 - Dec 2021

- Created a 3D third-person battleship shooter in Unity by modeling game assets, scripting player interactions in C#, and debugging gameplay mechanics.

ML & AI - Soybean Disease Identification

Dec 2021

- Implemented an image-based disease classification pipeline for soybeans using K-Nearest Neighbors and Convolutional Neural Networks, including data preprocessing, training, and evaluation in Python.

ColabNotebook:

https://colab.research.google.com/drive/1Cvho5C2_krwcidJnoMr6_mQUjRbUojM3?usp=share_link

IoT Indoor Mapping Remote-Controlled Car

Apr 2021 - May 2021

- Supervised the development of a networked RC car for indoor mapping by integrating Raspberry Pi 4B, Arduino Due, IMU sensors, and motor controllers, with synchronized control logic in Python and C++.

GitHub: <https://github.com/sf6472/CS370FinalProject>

Tourism Route Planning Web App

Jan 2021 - May 2021

- Crafted a dynamic web application for travel route planning using Java (Spark), React, SQL, and REST APIs, overseeing UI/UX flow, backend logic, and database interactions.

GitHub: <https://github.com/ChaoXiao26/CSU-CS314-Spring2021>

IoT Voice-Interactive Device

Dec 2020 - Mar 2021

- Delivered a voice-controlled IoT device with Adafruit Grand Central M4, Seeduino Xiao, and ESP8266 modules, integrating audio hardware and custom firmware.

Bluetooth Automatic Computer Switch

Apr 2018

- Assembled a Bluetooth-enabled computer power switch using Atmega328p and HM-10 BLE, programming control firmware in C++, and fabricating custom circuit hardware.

GitHub: <https://github.com/sf6472/BTComputerSwitch/>

PROFESSIONAL SKILLS

- Programming Languages:** C, C++, Python, Java, SQL
- Embedded Systems & Hardware:** Arduino, Raspberry Pi, ESP32, Sensors & Actuators, Hardware-Software Integration, Circuit Design, Soldering
- Databases:** SQL Database Integration
- Networking:** TCP/IP, Routers, Switches, Firewalls, Network Setup & Configuration
- Cloud Platforms:** AWS, Microsoft Azure, Google Cloud Platform (GCP)
- Tools & Environments:** Git, Linux, Visual Studio Code, EasyEDA
- Core Competencies:** Debugging, Troubleshooting, CCTV/Surveillance Deployment, Team Leadership, Training & Development, Stakeholder Management, Incident Management & Resolution, Process Optimization, Technical Documentation & Reporting, KPI-Driven Benchmarking, Prioritization, Cloud-Based Data Warehousing

EDUCATION

Colorado State University -Fort Collins, CO

Bachelor of Science in Computer Science

GPA: 3.17

Jul 2017 - Dec 2021

Relevant Coursework: Computer Organization, Software Development, Systems Security, Database Systems, Machine Learning, Artificial Intelligence

Electrical Engineering Coursework

DC Circuit Analysis, Problem Solving for Engineers

LANGUAGES

English, Mandarin